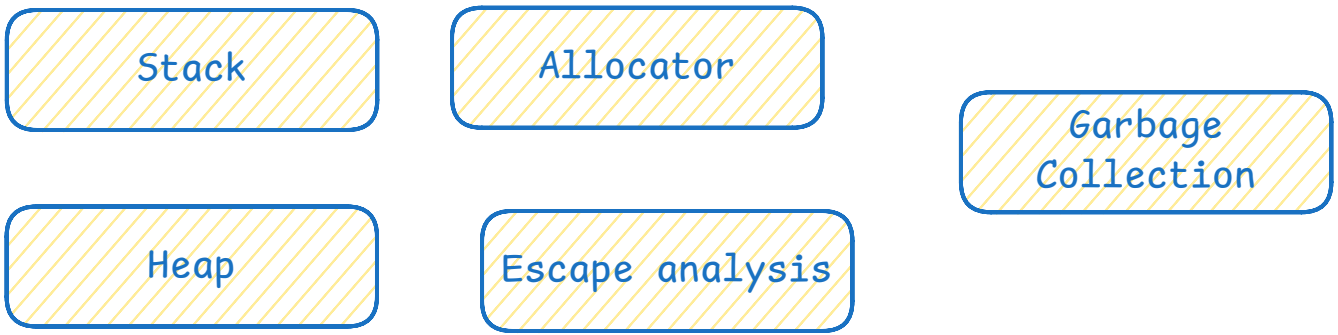
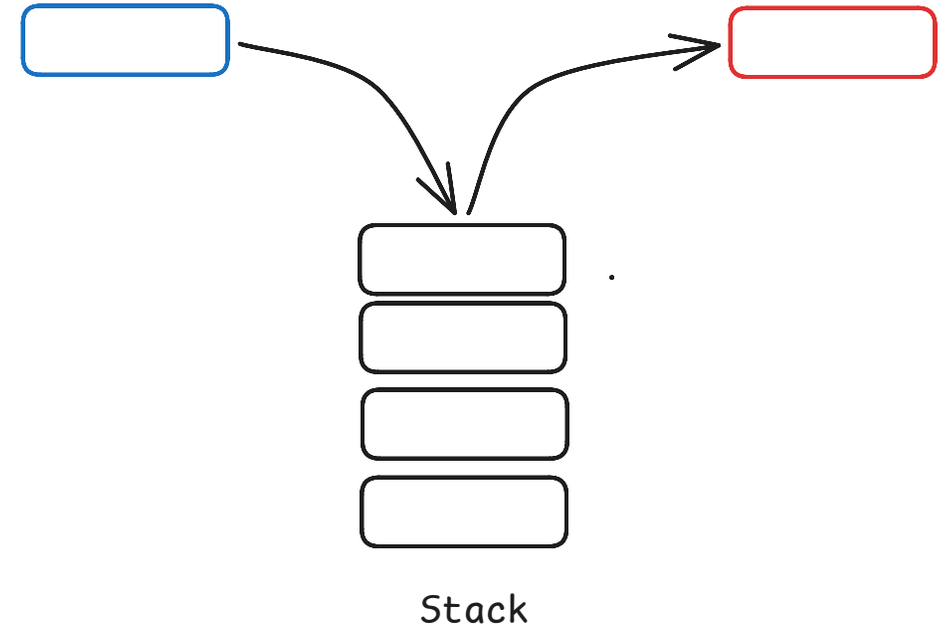
Lecture 07 - Memory Management



Memory Management

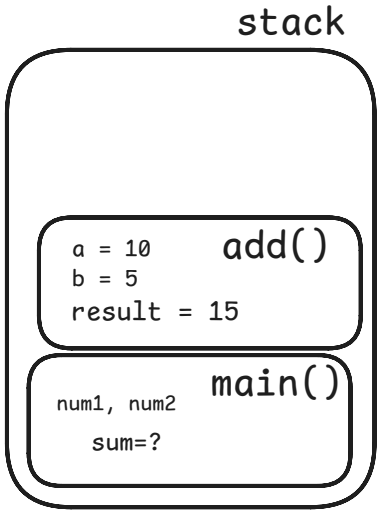
is the process of allocating, using, tracking, and reclaiming memory while a program is running.



Let's consider the following Go code:

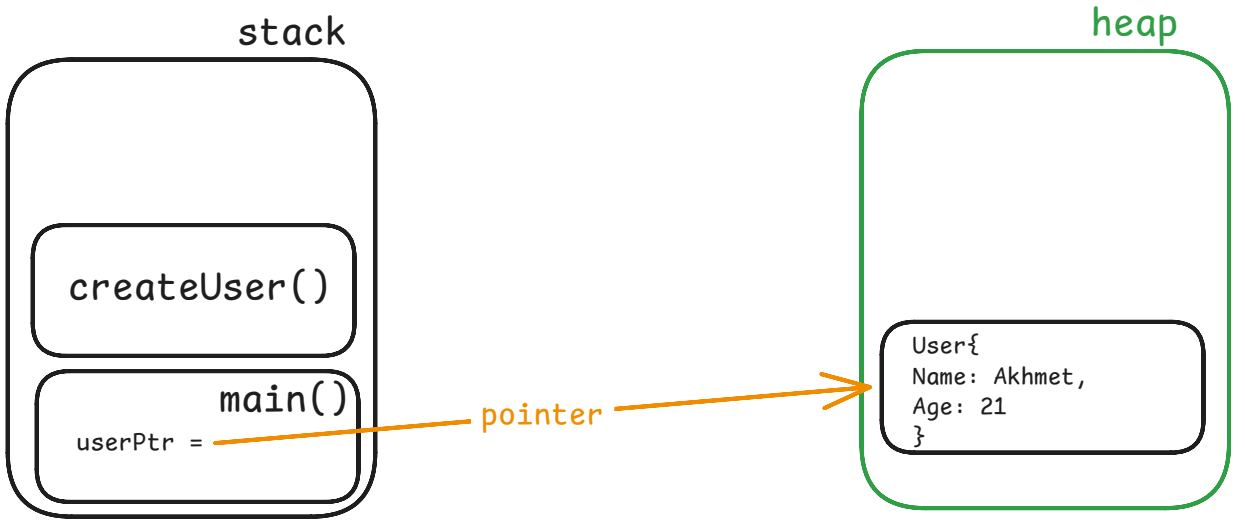
```
func main() {
    age := 25
    name := "Alice"

    numbers := make([]int, 100)

    fmt.Println(age, name, numbers)
}
```

```
func add(a, b int) int {
    result := a + b
    return result
}

func main() {
    num1 := 10
    num2 := 5
    sum := add(num1, num2)
}
```




```
go run .
```

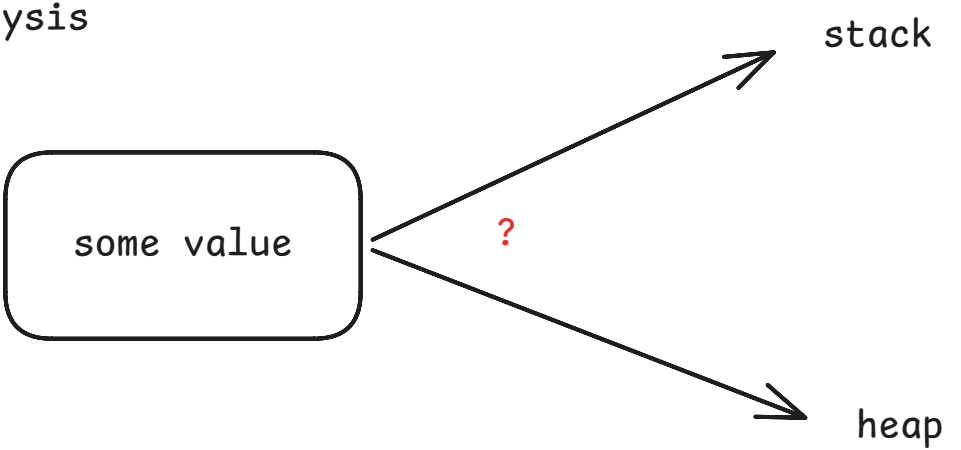
```go
func createUser() *User {
    user := User{
        Age: 21
        Name: Akhmet
    }

    return &user
}

func main() {
    userPtr := createUser()
}
```



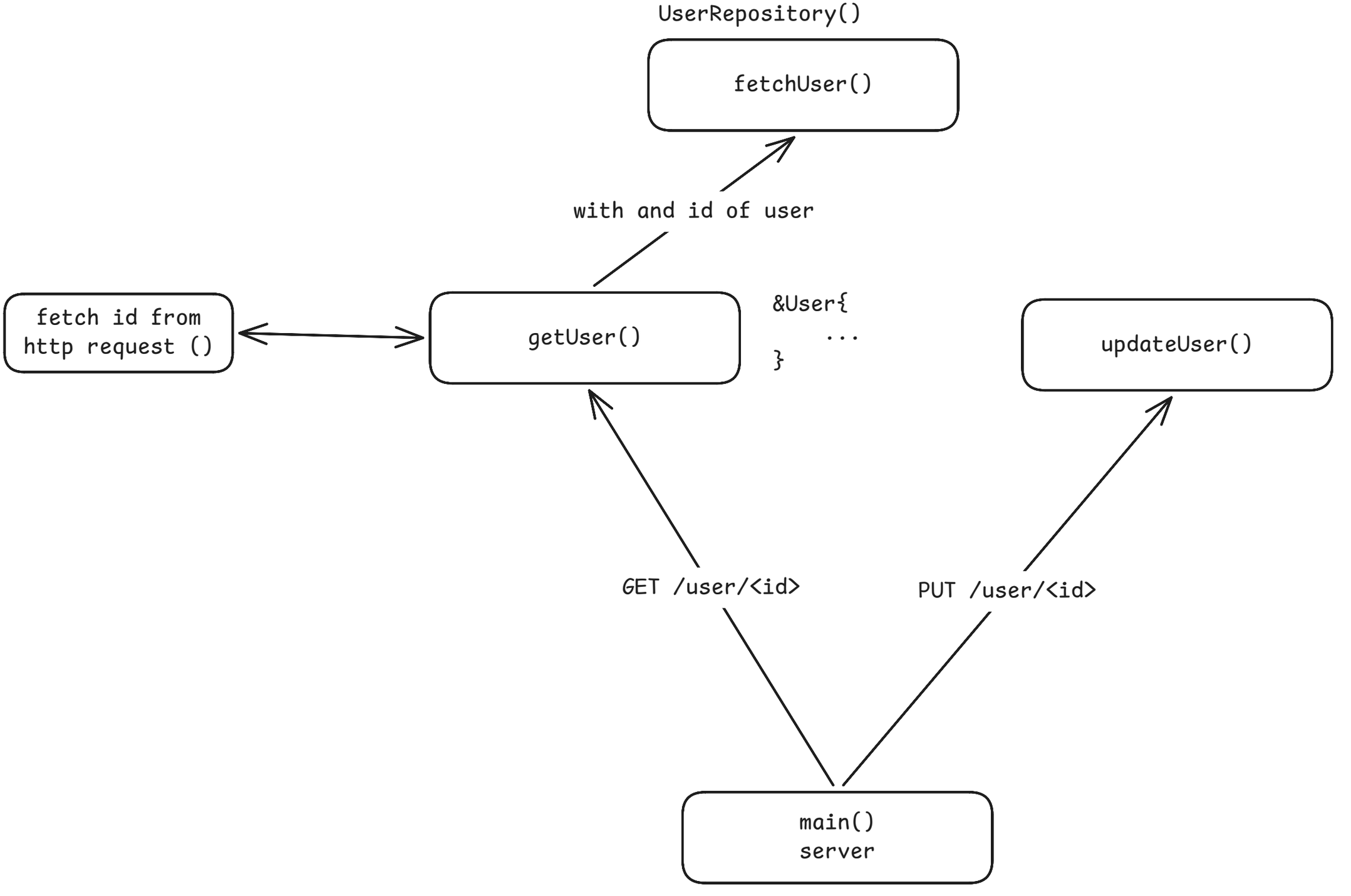
compiler: xmm, the user could be used further...

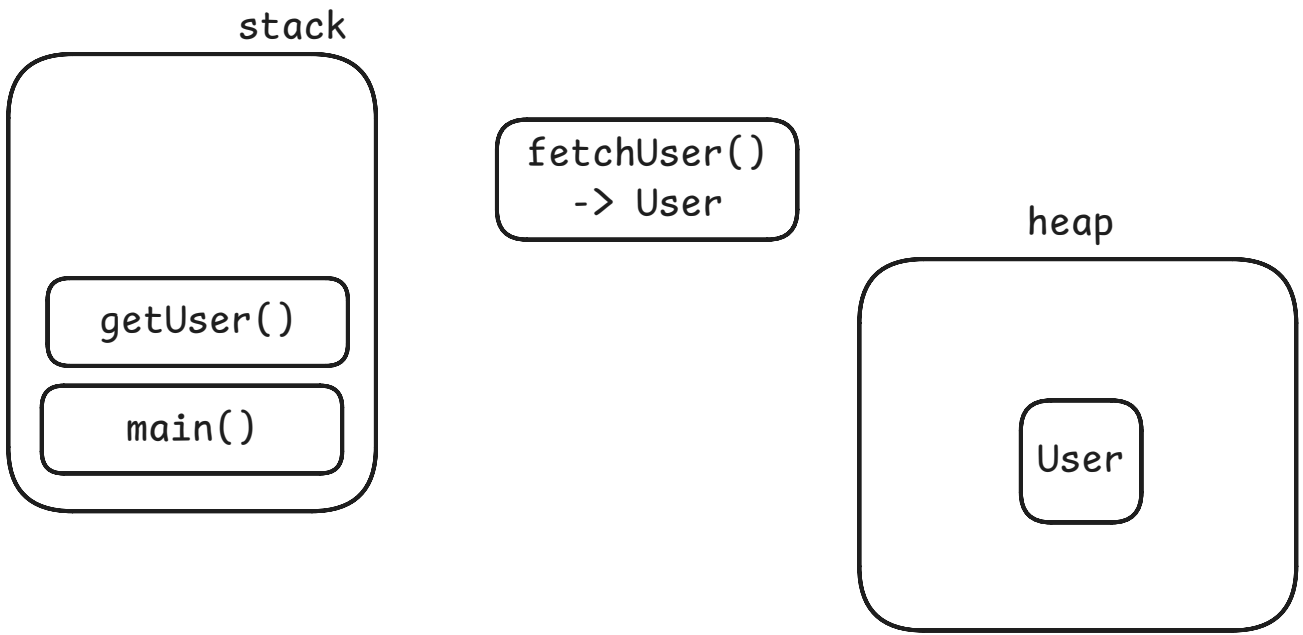
Using a pointer doesn't automatically cause heap allocation!

Escape analysis



```
var global *int


func save() {
    x := 100
    global = &x
}



 func double(n *int) {
     *n *= 2
 }

 func calculate() {
     x := 10
     double(&x)
     return x
 }
```


UserRepository()
fetchUser()
with and id of user
fetch id from
http request ()
getUser()
&User{
...
}
updateUser()
GET /user/<id>
PUT /user/<id>
main()
server


stack
getUser()
main()
fetchUser()
-> User
heap
User